ChatGPT

Executive Summary

Mediterranean seabed polymetallic deposits are largely undocumented: **no extensive manganese nodule fields have been found**, as high sedimentation and generally shallow basins (mostly <5 000 m) inhibit their growth【67†L18-L24】. **Co-rich ferromanganese crusts** are feasible on hard-rock highs (seamounts/volcanoes) of the Hellenic and Aegean arcs, where upwelling and volcanic substrate favour hydrogenetic crusts【102†L198-L206】. For example, the Ionian Sea's deepest trough (Calypso Deep, 36.567° N, 21.133° E) reaches ~5 200 m【117†L136-L140】 – within the global nodule depth range 【67†L18-L24】 – but is also heavily sedimented. In contrast, volcanic vents (e.g. near Santorini/Kolumbo, Milos, Nisyros) already host **hydrothermal polymetallic deposits** (Pb–Zn–Au, CO_2 seeps) with high metal grades and thus imply conditions for metal enrichment【123†L27-L36】【102†L198-L206】. We have identified priority target zones (Table 1) around Greece: (1) **Ionian abyssal basin (Calypso Deep, 36.567° N 21.133° E)** for nodules (depth ~5 100 m, highly condensed sediments); (2) **Hellenic Arc seamounts (Santorini/Kolumbo, 36.980° N 25.370° E)** for crusts/sulfides (depth ~1 500 m, volcanic bedrock); (3) **Milos–Cyclades region (36.420° N 25.260° E)** and (4) **Nisyros–Dodecanese (36.610° N 27.120° E)** for vent-associated metal deposit exploration. These are ranked by evidence of past volcanic/hydrothermal activity. **Assumptions:** Bathymetry from GEBCO/ETOPO; depths and locations use WGS84; data drawn from open literature and survey reports; environmental/regulatory issues (e.g. protected areas, Mediterranean dumping bans) are noted but not the focus.

【139†embed_image】 *Figure: Bathymetric map of the central–eastern Mediterranean (Aegean Sea shown) from GEBCO/ETOPO1【117†L136-L140】. Notable features: deep Ionian trough (top left) and Hellenic arc volcanoes (southern Aegean). Major troughs (Calypso 5 109 m) are labeled.*

Geology and Bathymetry of the Greek Mediterranean

The Mediterranean basin has complex tectonics: a narrow, young ocean between converging plates. West Mediterranean basins are relatively shallow, but the **Ionian Sea** hosts the Calypso Deep (~5 200 m) 【117†L136-L140】. Bathymetry (Fig. 1) shows that most Greek waters are <2–3 km deep (Aegean) except the Hellenic trench. Thick terrigenous sediments (from Africa, rivers) cover basins, especially Ionian (km-scale accumulations). By contrast, volcanic highs (Aegean arc, Cyclades islands, Dodecanese) stand above basement and have thin or no sediment cover. These factors govern mineralization: nodules require very slow sedimentation (as in abyssal plains), whereas **cobalt-rich crusts** form on exposed rocks under strong currents【102†L198-L206】【67†L18-L24】.

Polymetallic Nodules

Globally, manganese nodules occur on abyssal plains at ~3 500–6 000 m【67†L18-L24】 and contain ~29% Mn plus Ni, Cu, Co【77†L1-L9】【67†L18-L24】. In the Mediterranean, no extensive fields are reported. The Ionian Deep (Calypso) is within this depth range【117†L136-L140】 but carries >3 km of sediment from Africa. Sediment rates are too high for nodules to remain on the surface, so any nodules would be rare or buried. Shallower Mediterranean basins are outside the nodule zone. *Table 1* lists the sole true nodules target: Ionian Deep. Evidence is purely inferential (global analogy); confidence is **low**. **Next steps:** geophysical profiling and targeted coring on any sediment-starved flat at 4–5 km depth.

Cobalt-Rich Crusts and Related Deposits

Cobalt-rich ferromanganese **crusts** grow on seamounts/plateaus (400–5 000 m) where currents inhibit sediment【102†L198-L206】. They contain Co (up to ~2%), Ni, Mn, Cu, V, REEs and trace PGEs【67†L18-L24】【102†L198-L206】. In Greek waters, prospective crust hosts are submarine volcanoes along the Hellenic and Aegean arcs, and the undersea ridges (Fig. 1). **Santorini–Kolumbo (36.98 N, 25.37 E)** lies on the active arc; Kolumbo vent (1 500 m depth) has already yielded Zn–Pb–Au ores, indicating metal-rich fluids. **Milos area (36.42 N, 25.26 E)** contains multiple hydrothermal vent sites【123†L27-L36】 (gassing seeps, metal deposits) – where Fe–Mn crusts may also occur on flanks. **Nisyros (36.61 N, 27.12 E)** hosts the Avyssos Caldera vent【34†L114-L123】. These targets have hard basaltic substrates. No confirmed crust sample from Med seamounts is published, so evidence is **moderate** (analogy to Pacific/Atlantic seamount crusts【102†L198-L206】). **Next steps:** high-resolution bathymetry and magnetics to map seamount extents; ROV/rock-drill sampling to detect crust layers; chemical analysis for Co/Ni content (similar workflows as Pacific surveys【72†L142-L149】).

Geophysical and Sediment Data Sources

Bathymetry is mapped by GEBCO and EMODnet (see Fig. 1). Sediment distribution is from legacy echosounder and cores: deep basins (Ionian, Hellenic) are mud-dominated; shelf edges and arc slopes often carbonates. Gravity and magnetics surveys (e.g. IBCM-M) show thin crust under the Western Mediterranean and thinned/crystalline crust beneath the Aegean【107†L1-L9】【107†L27-L36】. These indicate locations of buried basins versus exposed volcanics. Seismic profiles (industry/academic) outline basement highs. Integrating these datasets can highlight sites with basement close to seafloor (good for crusts) and unusually low sediment cover (good for nodules).

Metal Composition and Resource Potential

Manganese nodules typically contain ~30–35% MnO_2, Ni (~1–1.5%), Cu (~0.5–1%), Co (~0.2%)【67†L18-L24】. Ferromanganese crusts are even richer in Co (often 0.5–2%) and contain ~20–40% MnO_2 plus Ni, Cu and trace REEs【67†L18-L24】【102†L198-L206】. No assays exist for Mediterranean seafloor samples, but regional basalts are Fe–Mn rich. Hydrothermal vents (Kolumbo, Avyssos) have much higher Cu/Pb/Zn and Au (up to tens of g/t Au in Kolumbo reports). We assume crusts would approach world averages (e.g. Co ~0.5–1.0%). However, without samples this is speculative; **confidence** in composition is low until sampling is done. Initial sampling programs should include XRF and ICP-MS analysis of crust/nodule material and sediments for metal content.

Target Ranking and Recommended Steps

The primary exploration targets (Table 1) are ranked by evidence: Hydrothermal deposits (Kolumbo, Avyssos) are highest confidence due to discovery. Crust-hosting seamounts (Milos area) follow. Abyssal plain nodules (Ionian) are lowest priority given low evidence. **Table 1** compares target coordinates, depths, deposit types, evidence level and next actions. For all targets, *first steps* include compiling multibeam bathymetry and gravimagnetic grids (available from EMODnet), followed by ship surveys (detailed mapping) and ROV/submersible sampling of seafloor rocks or sediments. Flowchart 1 (below) summarizes the recommended exploration workflow.

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flowchart LR
    A[Gather existing bathymetry & seismic data] --> B[Identify potential
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zones]
    B --> C{Potential target types}
    C -->|Nodules| D[Deep Ionian plain (calm sediment)]
    C -->|Crusts| E[Volcanic seamounts (Aegean arc)]
    C -->|Sulfides| F[Hydrothermal vents (Kolumbo, Nisyros)]
    D --> G[Assess sediment thickness & currents]
    E --> G
    F --> G
    G --> H[Field survey planning: MBES & magnetics]
    H --> I[ROV/submersible sampling of nodules/crust]
    H --> J[Drill cores from abyssal sediment]
    I --> K[Lab analysis (composition, age)]
    J --> K
    K --> L[Rank and prioritize targets]
    L --> M[Detailed sampling & resource evaluation]
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Figure: Proposed seabed exploration workflow: integrate geophysical data (multibeam bathymetry, magnetics) with geological analysis to define sampling targets; follow with ROV/drilling and geochemical assays to confirm deposits.

Legal/Environmental Notes (brief)

All Mediterranean seabed activities fall under national EEZ or regional agreements (Barcelona Convention). Greece's waters include Marine Protected Areas (e.g. Posidonia meadows) where mining is likely prohibited. Environmental impacts (habitat disturbance) are a concern. Relevant laws (e.g. EU Habitats Directive) would apply. We assume future work respects restrictions (e.g. avoid World Heritage sites).

Table 1. Comparison of polymetallic targets in Greek Mediterranean waters. Coordinates in WGS84. Evidence is classified by actual sampling or geological inference.

Target (Region)	Coordinates (Lat, Lon)	Depth (m)	Deposit Type	Evidence Level	Next Steps
Ionian Abyssal Plain	36.567° N, 21.133° E (Calypso)	~5 100	Mn nodules	Low (none found)	Geophysics (MBES), core sampling
Santorini–Kolumbo Seamount	36.980° N, 25.370° E	~1 500	Hydrothermal sulfides / crust	High (vent found)	ROV sampling of crust/vent
Milos–Cyclades seamounts	36.420° N, 25.260° E	800–1 000	Ferromanganese crust / vents	Moderate (new vents)	Geophysical mapping, ROV cores
Nisyros Caldera (Avyssos)	36.610° N, 27.120° E	~1 650	Hydrothermal sulfides	High (vent found)	Sample mineralized rocks/fluids

Sources: Abundance and depth ranges of nodules/crusts【67†L18-L24】【102†L198-L206】; Mediterranean bathymetry【117†L136-L140】; Greek marine surveys and hydrology【123†L27-L36】. (See references above.)